

Particle Physics  
Winter Quarter 2025  
Physics 130A



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Physics 317

**Lecture:** M,W,F 9:00-9:50 AM in Lecture Teaching and Learning Complex 3210 (next to Silo, t

**Textbooks:** G: Introduction to Elementary Particles (2nd), Griffiths.  
LN: Lecture Notes (Course website)  
T: Modern Particle Physics, Thomson. (Optional)  
An Introduction to Tensors and Group Theory for Physicists, Jeevanjee. (Optional)  
(Jeevanjee is available online at the library)

**Course TAs:** (TBD)

**Office Hours:** Mulhearn: (TBD)

**Homework:** From 5-10 homework problem sets.

**Midterm Exams:** Two In-Class Midterms

**Final Exam:** March, 21 2025 10:30 AM

**Course Description:**

Properties and classification of elementary particles and their interactions. Experimental techniques. Conservation laws and symmetries. Strong, electromagnetic, and weak interactions. Introduction to Feynman calculus.

**Reading:**

This course will be focused on Griffiths, and the lecture notes, which draw from Thomson and Jeevanjee. **You are expected to read Griffiths chapters one and two on your own.** That material is accessible and will be tested through multiple choice questions on the first midterm.

**Homework:**

There will be 5-10 homework assignments, which you will submit to gradescope. You may consult with other students but your submitted work must be your own. Your homework will be graded on (sincere) effort only. Homework will be a mixture of excellent problems from Griffiths and some custom problems designed to prepare you for the exams. The grading scheme for the course assumes students will be near 100% on homework. One late homework may be submitted at (or before) the final exam for full credit.

**Exams:**

Exams will be in class, paper and pen or pencil only, with no equation sheet, no internet, and no calculators. The exam will start with a quick answer, multiple choice part. After submitting that portion, you will pick up the second half of the exam, consisting of longer problems with some equations provided. No practice exams will be provided. Working through the homework problems for yourself will be best preparation for the exams. Timing and scope of the exams will be finalized during the quarter.

**Grades:**

Final grades will be a mixture of homework and exams. The weighting will be adjusted to avoid the need for a curve. It is assumed that your homework scores (graded on effort only) will be near 100%.

**Course Schedule:**

This is a preliminary schedule which will evolve as the course progresses.

You are expected to read and comprehend Griffith's Chapters 1 and 2 on your own, and that material will be included on the exams.

| <b>Week</b> | <b>Dates</b> | <b>Topics</b>                             | <b>Reading</b>            |
|-------------|--------------|-------------------------------------------|---------------------------|
| 1           | Jan 6,8,10   | Vectors and Tensors<br>Special Relativity | LN Relativity             |
| 2           | Jan 13,15,17 | Relativistic Kinematics                   | G 3.1-3.5                 |
| 3           | Jan 22,24    | Decay Rates and Cross Sections            | G: 6.1-6.2<br>TH: 3.1-3.5 |
| 4           | Jan 27,29,31 | Feynman Calculus                          | G: 6.3                    |
| 5           | Feb 3,5,7    | Symmetry and Spin                         | TBD                       |
| 6           | Feb 10,12,14 | Dirac Equation<br>Quantum Electrodynamics | G:7.1<br>G:7.2-7.3        |
| 7           | Feb 19,21    | Quantum Electrodynamics                   | G: 7.4-7.9                |
| 8           | Feb 24,26,28 | Quantum Chromodynamics                    | TBD                       |
| 9           | Mar 3,5,7    | TBD                                       |                           |
| 10          | Mar 10,12,14 | TBD                                       |                           |